

Artificial Intelligence — Complete Study Guide

A research-backed, intuition-first deep dive into AI — built for someone starting from zero and aiming to master it.

Your Learning Map

This guide is structured **chapter by chapter**, following your official syllabus. Each chapter file lives in the `chapters/` folder.

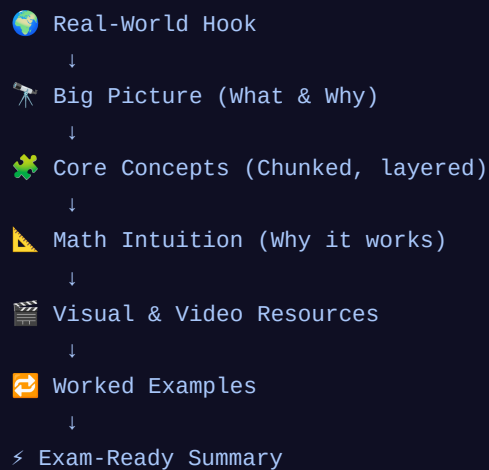
#	Chapter	Hours	Marks	Status
1	Introduction to AI	4 hrs	7	✓ Ready
2	Problem Solving	4 hrs	7	✓ Ready
3	Search Techniques	5 hrs	9	✓ Ready
4	Knowledge Representation & Reasoning	8 hrs	14	✓ Ready
5	Structured Knowledge Representation	4 hrs	7	✓ Ready
6	Machine Learning	6 hrs	10	✓ Ready
7	Applications of AI	14 hrs	26	✓ Ready

Total: 45 hrs → 80 marks exam

How to Use This Guide

Philosophy: First Principles → Intuition → Formalism

Every chapter follows this exact learning arc:



🎯 Recommended Study Order

Week 1: Ch 1 → Ch 2 (Foundation & Problem-Solving mindset)

Week 2: Ch 3 (Search — the heart of classical AI)

Week 3: Ch 4 + Ch 5 (Logic & Knowledge — the brain of AI)

Week 4: Ch 6 (Machine Learning — where AI learns)

Week 5: Ch 7 (Applications — where it all comes together)

🧰 Master Resource Toolkit

🎥 Must-Watch YouTube Channels

Channel	Best For	Link
3Blue1Brown	Neural networks, math intuition, visual proofs	youtube.com/@3blue1brown
StatQuest (Josh Starmer)	Bayes, statistics, ML algorithms clearly	youtube.com/@statquest
Sebastian Lague	Search algorithms visualized in games	youtube.com/@SebastianLague
Andrej Karpathy	Deep learning from first principles	youtube.com/@AndrejKarpathy

Interactive Visualizers

Tool	What It Shows
Red Blob Games	A*, BFS, DFS pathfinding — the BEST
PathFinding.js	All search algorithms side by side
Algorithm Visualizer	Code + animation side by side
NN Playground (TensorFlow)	Neural networks you can play with live

Reference Books (from syllabus)

Book	Why Read It
Russell & Norvig, <i>AI: A Modern Approach</i>	The Bible of AI — incredibly readable
E. Rich & Knight, <i>Artificial Intelligence</i>	Classic — great for exam preparation
P.H. Winston, <i>Artificial Intelligence</i>	First-principles approach, excellent for beginners



Exam Strategy

80 marks total. Here is where to focus:

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Chapter 7 (Applications) ————— 26 marks ← HIGHEST PRIORITY
Chapter 4 (Knowledge/Logic) ——— 14 marks ← SECOND PRIORITY
Chapter 3 (Search Techniques) — 9 marks
Chapter 6 (Machine Learning) — 10 marks
Chapters 1, 2, 5 ————— 21 marks ← Don't neglect!
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Start with Chapter 1 → `chapters/ch01_introduction.md`